

# AYUSH JUVEKAR

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## EDUCATION

**Michigan Technological University**, Houghton, MI  
*M.S. Computer Science*

**CGPA: 3.9/4.0**  
Graduated April 24, 2026

**Pune Institute of Computer Technology**, Pune, India  
*B.E. Computer Engineering*

**GPA: 8.14/10**  
June 2024

## TECHNICAL SKILLS

**Languages:** Python, JavaScript/TypeScript, Java, C++, Go, Rust, C# | **Frontend:** React, Next.js, TailwindCSS, shadcn/ui |  
**Backend:** Node.js, Express.js, Prisma ORM | **Databases:** PostgreSQL, MongoDB | **ML/AI:** TensorFlow, Scikit-learn,  
OpenCV, NumPy, Pandas | **DevOps:** Docker, Git, Vercel, Linux | **IoT:** Arduino, MQTT, Zigbee | **Auth:** Clerk, JWT

## PROJECTS

**YayCamp - Full-Stack Camping Platform** | *Next.js, TypeScript, Prisma, PostgreSQL* | yaycamp.ayushjuvekar.com

- Architected full-stack app with Next.js 14 SSR and Clerk authentication managing 100+ users, reducing page load time 40%
- Designed PostgreSQL schema with Prisma ORM and React Leaflet mapping, optimizing queries by 30% and achieving 99.9% uptime

**MonkeyLang-CS - Programming Language Interpreter** | *C#* | github.com/AyushJ1001/MonkeyLang-CS

- Implemented complete lexer, parser, and tree-walking interpreter supporting functions, closures, and first-class functions
- Built REPL environment with 95%+ test coverage across lexical analysis, parsing, and evaluation modules

**Anomaly Detection in Crowd Surveillance** | *Python, TensorFlow, Edge Computing* | CVBDSL Lab

- Trained CNN achieving 92% accuracy on 10K+ frames; optimized with TensorFlow Lite reducing inference time 73% on Jetson Nano
- Deployed real-time pipeline processing 720p at 25 FPS across 5 edge devices, cutting cloud costs 85% and response time 40%

**Particle Vibration Analysis System** | *Python, OpenCV, ZED Cameras* | Michigan Tech Research

- Developed CV system tracking 50+ particles at 60 FPS with sub-mm precision using stereo cameras and 3D data pipeline
- Applied statistical analysis correlating motion patterns with vibration frequencies, reducing manual analysis time 75%

**Secure Data Analytics with Homomorphic Encryption** | *Python, CKKS, Docker* | CVBDSL Lab

- Built privacy-preserving pipeline using CKKS encryption for healthcare data, achieving 98% accuracy with HIPAA compliance
- Containerized modules with Docker, streamlining deployment and improving processing efficiency 25%

**IoT Air Quality Monitoring System** | *C++, Arduino, MQTT, Python* | **Published: ESCI 2023**

- Engineered CO monitoring system with MQTT protocol, reducing detection time 35% via ML predictive alerts
- Presented at ESCI 2023 Int'l Conference; improved accuracy 20% through Kalman filtering and Zigbee mesh network

**Ranking App - Tournament Bracket System** | *TypeScript, React, Redis* | ranking.ayushjuvekar.com

- Created bracket app with decision-time-weighted scoring algorithm; implemented real-time Redis persistence for multi-device sync

## EXPERIENCE

**Teaching Assistant - Physics Department** | Michigan Technological University

Sep 2024 - Apr 24, 2026

- Facilitate learning for 60+ students in PH 2100/2200; developed Python automation reducing admin work 20 hours/semester
- Conduct tutorials improving average exam scores 15%; collaborate with faculty on course materials and assessments

## PUBLICATIONS

A. Juvekar et al., "Carbon Monoxide Concentration Monitoring System," *ESCI 2023*, DOI: 10.1109/ESCI56872.2023.10100144

## AWARDS

**2nd Place PICT InC 24** (Data Structures) | **Conference Presenter ESCI 2023** | **Mathex & MSCE Scholarships**  
2021-2024